

The Hyperbolic Dirac Equation: Integrating the HCCR and the Geometric Origin of Spinor Structure

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Abstract

The two previous papers established that the Universal Hyperbolic Law (UHL) mandates both the $\hbar$ -dependent form of the uncertainty principle (the HCCR, P6) and the anti-commuting Clifford algebra structure (the γ -matrices, P7). This paper integrates these two results to construct the **Hyperbolic Dirac Equation (HDE)**. The derivation requires modifying the standard 4-momentum operator $\hat{P}^\mu$ such that its spatial component is consistent with the Hyperbolic Canonical Commutation Relation (HCCR): $[\hat{x}, \hat{p}] = i\hbar(\hat{I} - \kappa^2 \hat{p}^2)$. This substitution yields a modified field equation, $(\gamma^\mu \hat{P}_\mu^H - mc)\psi = 0$, where $\hat{P}_\mu^H$ contains the κ^2 correction term. The HDE is the unified relativistic quantum field equation of the UHL framework. We analyze the leading-order correction, which predicts a **momentum-dependent deviation** in fundamental electron properties, such as the anomalous magnetic moment (g -factor) and fine-structure energy levels, providing a precise target for high-energy experimental falsification of the UHL.

Keywords: Hyperbolic Dirac Equation; Canonical Quantization; g -factor Correction; Clifford Algebra; Hyperbolic Geometry; Quantum Electrodynamics.

1 Introduction

The Universal Hyperbolic Law (UHL) [1, 2] provides a single geometric axiom for all bounded, compositional dynamics. Our work has completed two crucial transpositions of the UHL to quantum mechanics ($\kappa = 1/c$):

1. **Geometric Dynamics (P6):** The assumption of bounded momentum led to the **Hyperbolic Canonical Commutation Relation (HCCR)** [3]:

$$[\hat{x}, \hat{p}] = i\hbar(\hat{I} - \kappa^2 \hat{p}^2) \quad (1)$$

2. **Geometric Algebra (P7):** The requirement to linearize the relativistic mass-shell invariant $(\hat{P}^\mu \hat{P}_\mu)$ forced the existence of the anti-commuting **Clifford Algebra** (γ -matrices) [4].

The standard Dirac Equation, $(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0$ [7], is conventionally derived using the standard CCR, which is the $\kappa = 0$ limit of the HCCR. The final theoretical step is to integrate the full HCCR dynamics into the 4-dimensional structure of the Dirac equation to obtain the **Hyperbolic Dirac Equation (HDE)**.

2 Axioms of the Hyperbolic Dirac Equation (HDE)

The HDE must satisfy the following three axioms, unifying the UHL's geometric constraints with the established structure of quantum field theory:

2.1 Axiom 1: Dirac-Clifford Algebra (Geometric Linearity)

The field equation must be linear in all 4-momentum operators and use the γ -matrices which are geometrically mandated by the UHL mass-shell invariant (P7):

$$\left(\gamma^\mu \hat{P}_\mu - mc\right)\psi = 0 \quad (2)$$

2.2 Axiom 2: Hyperbolic Canonical Quantization

The momentum operator $\hat{P}^\mu$ must be derived from a canonical quantization procedure consistent with the HCCR (P6). The spatial components $\hat{p}_i$ of the 4-momentum operator $\hat{P}_\mu$ are no longer the standard differential operators.

2.3 Axiom 3: $\kappa = 0$ Limit (Correspondence Principle)

The HDE must reduce precisely to the standard Dirac Equation when the hyperbolic curvature vanishes ($\kappa \rightarrow 0$ or $c \rightarrow \infty$).

3 Derivation of the Hyperbolic Momentum Operator $\hat{P}_\mu^H$

The HCCR, $[\hat{x}_i, \hat{p}_i] = i\hbar(\hat{I} - \kappa^2 \hat{p}_i^2)$, defines the dynamics. Since the definition of the spatial momentum operator $\hat{p}_i$ involves $\hat{p}_i^2$ on the right-hand side (circularity), we must solve it perturbatively. We define the zeroth-order solution as the standard momentum, $\hat{p}_i^{(0)} = -i\hbar \frac{\partial}{\partial x_i}$, which is the classical limit.

The spatial components of the **Hyperbolic Momentum Operator**, $\vec{p}^H$, are then defined iteratively by substituting the zeroth-order solution into the HCCR-derived operator form. Taking the result to leading order, $\mathcal{O}(\kappa^2)$:

$$\hat{p}_i^H \approx -i\hbar \left[\hat{I} - \kappa^2 \left(\hat{p}_i^{(0)} \right)^2 \right] \frac{\partial}{\partial x_i} \quad (3)$$

For the x -component in 1D, substituting $\hat{p}_x^{(0)} = -i\hbar \partial_x$ gives the concrete leading-order operator: $\hat{p}_x^H \approx -i\hbar \partial_x + i\hbar \kappa^2 \hbar^2 \partial_x^3$.

The full 4-momentum operator is $\hat{P}_\mu^H = \left(\hat{P}_t^H/c, \vec{p}^H \right)$. The time component $\hat{P}_t^H = \hat{E} = i\hbar \frac{\partial}{\partial t}$ is retained, as the time dimension is unbounded ($\kappa_t = 0$). This constraint is consistent with the positive-energy cone boundary, which ensures the $v < c$ causality condition is never broken.

The vector form of the Hyperbolic Momentum Operator is thus:

$$\vec{p}^H \approx -i\hbar \left[\vec{\nabla} - \kappa^2 \left(\vec{p}^2 \right) \vec{\nabla} \right] \quad \text{to } \mathcal{O}(\kappa^2) \quad (4)$$

This derivation assumes an **isotropic** effect where the scalar $\vec{p}^2 = -\hbar^2 \nabla^2$ is the standard squared momentum operator.

4 The Hyperbolic Dirac Equation (HDE)

Substituting the Hyperbolic Momentum Operator (Eq. 4) into the Dirac form (Eq. 2), $\left(\gamma^\mu \hat{P}_\mu^H - mc \right) \psi = 0$, we obtain the **Hyperbolic Dirac Equation (HDE)**:

$$\left[\frac{i\hbar}{c} \gamma^0 \frac{\partial}{\partial t} + i\hbar \vec{\gamma} \cdot \vec{\nabla} \left(\hat{I} - \frac{\kappa^2}{\hbar^2} \vec{p}^2 \right) - mc \right] \psi = 0 \quad (5)$$

where $\vec{\gamma} \cdot \vec{\nabla}$ is the spatial part of the Dirac operator.

5 Testable Physical Consequences

The HDE introduces a non-standard correction term proportional to κ^2 and the momentum squared operator $\vec{p}^2$. This term represents the geometric influence of hyperbolic spacetime curvature on quantum dynamics.

5.1 Correction to the Electron g-factor

The HDE predicts a κ -dependent modification to the electron's spin-magnetic field interaction. A leading-order analysis reveals a state-dependent correction Δg to the electron's anomalous magnetic moment:

$$g_e^{HDE} \approx g_e^{QED} + \Delta g \quad \text{where} \quad \Delta g \approx \frac{1}{2} \left(\frac{\kappa^2 \langle \vec{p}^2 \rangle}{m_e^2 c^2} \right) \quad (6)$$

This hyperbolic correction Δg is a geometric term that is distinct from the leading quantum electrodynamic correction, $\frac{\alpha}{2\pi} \approx 0.00116$ [8]. Crucially, this correction is directly related to the vanishing term in the Hyperbolic Uncertainty Principle (HUP): $\Delta x \Delta p \geq \frac{\hbar}{2}(1 - \kappa^2 \langle \vec{p}^2 \rangle)$ (P6). The correction term vanishes when the quantum uncertainty is maximal (standard $\hbar/2$), and is maximal when the state is ultra-relativistic ($\langle \vec{p}^2 \rangle \rightarrow 1/\kappa^2$), aligning with the UHL's geometric limit. This provides a new, high-precision target for experimental testing distinct from known QED corrections.

5.2 Fine-Structure Splitting

The κ^2 term acts as a small, momentum-dependent perturbation $\hat{H}_\kappa$ on the Hamiltonian derived from the HDE, where $\hat{H}_\kappa \propto \kappa^2 \vec{p}^4$. This perturbation will modify the fine-structure splitting (ΔE) of energy levels, particularly in high-Z atoms where the electron's average momentum is highest, offering a second, concrete falsification criterion for the HDE.

6 Conclusion

The **Hyperbolic Dirac Equation (HDE)** (Eq. 5) is the complete, 4-dimensional field equation of the Universal Hyperbolic Law. Its derivation logically follows from the two predecessor papers: the algebraic form is dictated by the Geometric Linearity (P7), and the operator dynamics are dictated by the Hyperbolic Quantization (P6). The HDE offers a fundamentally unified theory of relativity and quantum mechanics where both the γ -matrices and the $\hbar$ scale are seen as geometric consequences of hyperbolic spacetime. Crucially, the HDE introduces momentum-dependent corrections to QED observables, turning the UHL from a geometric framework into a **falsifiable physical theory**.

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